Basin Entry and the Osterwalder–Schrader Frontier
Target 58 of the H1 Downstream Programme

Yang–Mills representational dynamics research programme

21 July 2026

Abstract

We audit the complete five-target run against physical basin entry and the Clay mass-gap target. A positive temporal-transfer countermodel proves that a fixed one-scale spatial defect ceiling is compatible with arbitrarily small temporal gaps. A cutoff family further shows that positive regulated gaps may vanish in physical units. The finite programme reaches C2, with C3 conditional bridges; the first physical nonlinear kernel inequality remains open.

1 What the five targets prove

The run establishes: (i) exact finite normalized disintegration; (ii) a finite Harnack criterion for inverse-vacuum moments; (iii) robust multichannel Perron envelopes; (iv) changing-state-space weight cocycles and the correct physical-length criterion; and (v) the present frontier audit. None of the finite witnesses is asserted to be the complete nonlinear $SU(2)$ measure.

2 Temporal independence countermodel

For $\varepsilon > 0$, define

$$T_\varepsilon = \frac{1}{2}\begin{pmatrix} 1+e^{-\varepsilon} & 1-e^{-\varepsilon} \\ 1-e^{-\varepsilon} & 1+e^{-\varepsilon} \end{pmatrix}.$$

It is symmetric, entrywise positive, and has eigenvalues 1 and $e^{-\varepsilon}$.

Proposition 1. *The transfer gap at unit time is ε. Therefore the same spatial weighted defect ceiling*

$$q = 0.002430277501841998$$

is compatible with temporal gaps tending to zero.

Proof. The transfer-matrix gap is $-\log(e^{-\varepsilon}/1) = \varepsilon$. The spatial defect kernel is an independent tensor factor and is unchanged as $\varepsilon \downarrow 0$. □

There is also a cutoff version. Let the temporal lattice spacing be a and choose $\varepsilon(a) = a^2$. Every regulator has a positive transfer gap, but the gap in physical units is

$$\frac{\varepsilon(a)}{a} = a \longrightarrow 0.$$

At $a = 2^{-12}$ the regulated physical gap is $2^{-12} = 0.000244140625$.

3 Gate audit

Obligation	Status
Complete normalized nonlinear $SU(2)$ retained kernel	open
Uniform vacuum/Lyapunov weight and root moment	open
Complete multichannel weighted ceiling below threshold	open
Repeated contraction in accumulated physical length	open
Continuum OS construction and reflection positivity	open
Dense state-separating temporal observable core	open
Identity with nontrivial pure Yang–Mills	open
Every compact simple gauge group	open

The first failure is the physical normalized kernel. The current claim level is C2, not C4. The conditional statement that dense-core temporal decay in an already constructed OS theory excludes spectrum in $(0, m)$ is a C3 bridge, not a construction theorem.

4 Limit order and next theorem

The defensible order is: construct finite normalized rows; prove uniform thermodynamic control; prove cutoff-uniform physical-length contraction; construct and identify the OS continuum theory; prove density of the decaying observable core; then apply the spectral-measure implication.

The first open physical target is

$$\sup_{a,L,h} \frac{(Q_{a,L}V_{a,L})(h)}{V_{a,L}(h)} < q_{\mathrm{crit}}, \qquad \sup_{a,L} \nu_{a,L}(V_{a,L}) < \infty,$$

for every positive-mass history and a weight defined without clustering, confinement, or a pre-existing gap.

Warning 1. *A one-scale spatial certificate, finite-volume transfer gap, formal RG flow, or assumed confinement scale does not establish the Hamiltonian mass gap. Continuum existence, OS axioms, observable density, theory identity, and physical-unit uniformity are separate obligations.*

5 Conclusion

The five targets improve the representation calculus and eliminate two major shortcuts. They do not solve Yang–Mills. Eleven audit tests verify the temporal spectra, cutoff collapse, claim-level gates, and conditional OS handoff.